Ultra-stable lasers and photonic integrated circuits for secure communications

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A dissertation submitted in partial fulfillment

of the requirements for the degree of

Master of Research — Centre for Doctoral Training

in

Electronic and Photonic Systems.

Word count: **9978**

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Part A: General declaration

I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

Signed: Joseph Pereira

Date: August 14, 2026

Part B: Declaration of contribution from other sources

Data: Experimental data from multiple publications was obtained to benchmark computational results. This was used by the author to plot figures, and is clearly stated in figure captions. All other data was generated by the user.

Software code: The system of rate equations was coded manually by the author, based on the framework defined by Gioannini and Rossetti (2011). The programme was troubleshooted with the assistance of generative AI model Gemini Pro v3.1. Gemini Pro v3.1 was also used to generalise helper functions for the PICWave-Python bridge, where initial code was written by the author. The functionality of the edits was manually checked. The script to generate parameter combinations using LHS was written by Gemini Pro v3.1- the author confirmed manually that sampling across the parameter space took place. The GP script was written by the author, although Gemini Pro v3.1 was used to help choose a kernel.

AI: Generative AI model Claude Sonnet 5 was used to find relevant papers in the literature review, but the papers were read and interpreted by the author. Claude Sonnet 5 was used to help edit this report by identifying areas to make more concise. All edits were performed manually by the author. Claude Sonnet 5 was used to help format graphs by writing matplotlib helper functions.

Abstract

Improvements in quantum computing threaten the security of current public key encryption methods. Continuous variable quantum key distribution (CV-QKD) is a method that is theoretically secure against an eavesdropper, and could make use of existing telecom hardware. One of the biggest issues facing QKD is the phase drift between two separate lasers used to send signals, which must be carefully controlled to reliably communicate.

With the aim of fabricating an on chip laser for quantum key distribution, the laser must be integrable on silicon and ultra-stable (low linewidth). In tightly controlled laboratory conditions quantum dot lasers satisfy both of these requirements. For a real device, however, the behaviour of these lasers in non-ideal conditions is crucial, but most models for simulating quantum dot lasers do not consider these effects.

In this work a simulation framework was developed to encode the complex physics of a quantum dot laser model in a photonic circuit design software (PICWave). The laser dynamics qualitatively agree with experimental quantum dot results, and a free-running linewidth of 160 kHz was calculated. This is comparable to experimental values. The flexibility of this model was demonstrated by simulating a quantum dot laser under external optical feedback. The model captures the high feedback resilience of quantum dot lasers, demonstrating linewidth reduction to as low as 39 Hz. This model therefore forms a basis to further investigate quantum dot laser behaviour in photonic circuits, and under non-ideal conditions.

Acknowledgements

I would like to thank my supervisor, Prof. Wale for his technical advice in our weekly meetings, pointers towards helpful research, and guidance of the project direction. I would also like to my secondary supervisor, Dr. Fice for the useful discussions on results, and along with Prof. Wale, his advice when writing this report. Thank you also to Dr. Ponnampalam for her advice on using the PICWave software, and Liv Hawkins for our discussion on quantum dot simulation.

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Chapter 1

Introduction

Advancements in quantum computing threaten the security of public key encryption, which underpins modern communication. Quantum key distribution (QKD) allows a key to be shared between two parties, for use in one-time pad encryption. By the laws of quantum physics, this technique is theoretically secure against an eavesdropper, hence has received significant attention in recent years.

QKD can be experimentally achieved via two methods - discrete variable (DV) or continuous variable (CV). Whereas DV QKD uses single photons, CV uses coherent states and coherent detection. CV QKD can offer a greater rate of data transfer[1], and are better suited to existing telecom hardware infrastructure [2], hence CV is the focus of this report. However, development of CV QKD been limited by the size, cost, and stability of existing laser technologies. The first two issues can be addressed by pivoting to photonic integrated circuits, which reduces the size and could reduce costs of the system. A prototype of such an integrated device has been demonstrated by Fatkhiev et al. [3]. However, this work uses an external oscillator, achieving a secret key rate (SKR) of 1.9 Mbit/s. An on-chip ultra-stable laser is the ultimate goal to build a fully-integrated, high SKR device.

A key metric to quantify laser phase stability is the linewidth. The maximum laser linewidth for CV-QKD has been estimated at 200 kHz. Quantum dot (QD) lasers can offer free-running linewidths below this value, and can be further reduced under external optical feedback (EOF). QD lasers have recently been grown epitaxially on Silicon, with UCL researchers demonstrating a first of its kind QD distributed feedback laser [4]. Therefore ultra-stable QD lasers could be integrated on chip. However, building a practical device with this technology requires overcoming several engineering challenges. A commercial device will experience a wider range of conditions than a laser in a laboratory environment. Although ultra-low linewidths can be achieved, this requires very high quality QD growth, which may not be reproducible at scale. This calls for a simulation framework which could investigate the behaviour of a QD laser under non-ideal conditions.

The charge dynamics in QD active media are complex, and this complexity is elevated when external factors are considered. QD lasers behave differently from existing quantum well lasers, owing to their discrete energy levels. Although QD models have been suggested in literature, they do not typically consider the effect of feedback. Additionally, the source code to model QD systems is not openly available. To model external factors, encoding QD laser behaviour into software which considers coupled physical effects is a natural choice. This report describes a simulation framework for simulating QD lasers, towards simulating external noise sources and temperature fluctuations. Such a software package could inform design parameters for QD lasers.

Throughout the design process and evaluation of the model, three specific criteria were considered:

1. Accuracy: QD lasers are complex systems, and approximations must be made. The model should at least qualitatively follow the same trends as experimental work.

2. Interpretability: The effect of tuning model parameters should be understood. This will help to explain why the model is/ is not accurate.

3. Flexibility: There should be scope to integrate the laser cavity in more complicated circuits, such that the effect of locking can be observed.

This report will explain why CV-QKD demands ultra-stable lasers, and why QD lasers are suitable for this purpose. The literature surrounding fabrication and performance of QD lasers will be reviewed, and used to develop a model to simulate QDs. The final model will be judged against the above criteria, followed by recommendations to improve the model.

Chapter 2

Theory

To understand the relevance of ultra-stable lasers to QKD, a basic understanding of the system is necessary. This section will investigate what phase noise is, and what the linewidth of a laser means. We will then define the requirements for linewidth in a QKD system. This will be followed by the methods of measuring linewidth, and the factors that influence it.

2.0.1 A brief introduction to QKD

In CV-QKD, Alice encodes a key via Gaussian modulation of the amplitude and phase of light, and sends it through an optical channel to Bob. This report will consider the ”local local oscillator” (LLO) implementation, where Bob makes a measurement using a local oscillator. The biggest challenge facing LLO quantum coherent communication protocols is phase reference sharing. The receiver must perform phase-sensitive detection using the local oscillator, whose phase drift must be controlled, estimated or corrected for. A shift in phase imparts random rotations on the quadrature basis of the sent data, manifesting as noise. The level of phase noise is therefore crucial to ensure that data can be sent securely at a fast rate [5].

2.0.2 Phase noise

Having established why phase noise is important in CV-QKD, this section will explain factors that cause phase noise. A phase shift can be introduced by intrinsic spontaneous emission events, environmental fluctuations, and hardware factors. The timescale over which these noise sources impact the phase of a laser varies. Some level of phase noise is unavoidable, and defined by amplified spontaneous emission (ASE) noise.

Spontaneous emission of a photon introduces an immediate phase shift (ϕ_i') and intensity

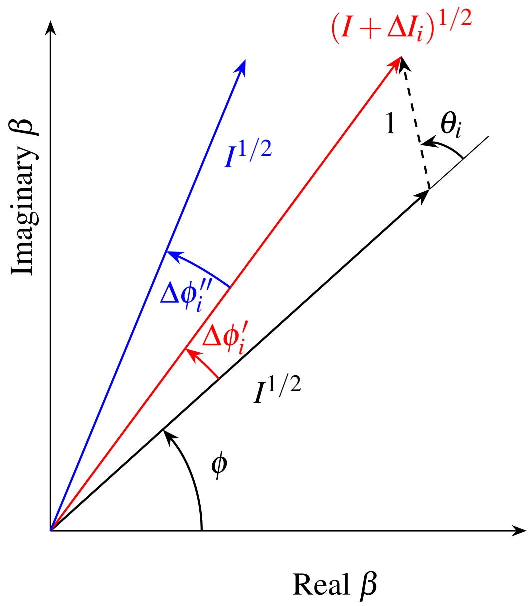

Figure 2.1: The instantaneous change of phase caused by the i^{th} spontaneous emission event is given as $\Delta\phi'$, and the electric field shown in red. The delayed change caused by intensity fluctuation is given as $\Delta\phi''$, and shown in blue.

shift ($\Delta I^{1/2}$) to the optical field. This is illustrated in Figure 2.1, shown in red.

However, there is an additional phase shift caused by the intensity change. When the intensity of the optical field increases, the laser undergoes relaxation oscillations, before returning to its steady state intensity. Whilst the optical field is elevated, there is a change in gain (caused by a net change in carrier density). This alters the refractive index, which introduces an additional, delayed phase shift, ϕ_i'' . The resulting optical field is shown in blue, with a total phase shift of $\phi_i' + \phi_i''$. This phase shift depends on the coupling between intensity and phase fluctuations, and is described by the linewidth enhancement factor (LEF or α_H) [6]:

$$\alpha_H = -\frac{4\pi}{\lambda}\frac{dn/dN}{dg/dN}. \tag{2.1}$$

It should be noted that the α_H parameter is difficult to define above threshold, because the gain clamps. As a result, the denominator in the above equation tends to 0, and becomes highly sensitive to noise. Above threshold an effective LEF may be defined, α_{eff}. This can be measured by applying a small signal current modulation. The time-domain intensity and frequency variations are measured. The Fourier Transform of these quantities describes the small signal response as a function of modulation frequency, termed as IM (intensity modulation) and FM (frequency modulation) responses. The effective LEF as a function of IM and FM is given as [7]:

$$\alpha_{eff}(f) = 2\frac{FM(f)}{IM(f)}. \tag{2.2}$$

The α_{eff} value plateaus as the modulation frequency increases, and this value is taken as the effective LEF.

2.0.3 What is the laser linewidth?

When ASE is the dominant noise contribution, the phase noise follows a random walk (scaling as $S_{\phi}(f) \propto f^{-2}$), and the resulting power spectral density (PSD) has a Lorentzian lineshape. The intrinsic linewidth is defined as the full width at half maximum (FWHM) of this Lorentzian lineshape, and is therefore a key value which quantifies laser phase noise. The intrinsic linewidth may be described by the Schawlow-Townes-Henry formula, given as [6]:

$$\Delta f = \frac{v_g^2 \cdot h\nu \cdot g \cdot n_{\mathrm{sp}} \alpha_m (1 + \alpha_H^2)}{8\pi P_{out}}, \tag{2.3}$$

where v_g denotes the group velocity,$h\nu$ the photon energy, g the gain, n_{sp} the spontaneous emission factor, α_m the mirror loss,P_{out} is the output power, and α_H the dimensionless linewidth enhancement factor. Therefore, along with the level of spontaneous emission, the linewidth enhancement factor of a material is an important determiner of the phase noise.

2.0.4 Linewidth requirements for CV-QKD

Now that the phase noise has been explained, its numerical effect on CV-QKD will be quantified. A key metric in QKD is the secret key rate, which measures the rate at which data can be securely transferred [1]. Any noise introduced by phase drift is conservatively attributed to an eavesdropper. Therefore as the phase noise increases, the secret key rate falls. Ultimately the phase noise must also be below the threshold set by security proofs. Marie et al. suggest a maximum laser linewidth of 200 kHz [5].

2.0.5 Frequency Noise

To extract the linewidth it is convenient to study the frequency noise of a laser, which is closely related to the phase noise. To briefly explain frequency noise, we first define frequency excursion as the difference between its temporal mean frequency and its instantaneous frequency:

$$\Delta\nu = \overline{\nu} - \nu(t). \tag{2.4}$$

The frequency noise, $S_\nu(f)$, is defined as the PSD of the frequency excursion. It is related to the phase noise by $S_\phi(f) = \frac{S_\nu(f)}{f^2}$. A random walk in phase noise (which determines the Lorentzian linewidth) therefore corresponds to a flat frequency noise. ASE and technical noise show different noise characteristics, and occupy different noise frequency regimes. Technical noise occurs at lower frequencies, where the noise power scales as f^{-1} (flicker/pink noise) or f^{-2} (random walk/red noise). ASE is a white noise which occurs at higher frequencies, and has a flat profile, f^0. These 3 regions are illustrated in Figure 2.2.

The intrinsic linewidth of the laser can be calculated from the flat region of the frequency noise spectrum. The intrinsic linewidth is calculated as

$$Intrinsic\ Linewidth = \pi S_{\nu,0}, \tag{2.5}$$

where $S_{\nu,0}$ is the level of intrinsic (white) frequency noise. A thorough derivation is covered by Jones [8].

2.0.6 Technical noise

Technical noise is made up of random walk noise - following f^{-2} - and flicker noise - following f^{-1}. Since this noise is not intrinsic to the laser system, it is not often included in QD laser simulations. Random walk frequency noise is attributed to environmental perturbations, for example fluctuations in temperature. Flicker noise is electronic in origin, for example from noise in signals from electronic drivers. There is no analytical form for the PSD of the technical noise, but it has been suggested that technical noise can be approximated by a Gaussian spectrum [10].

2.1 Quantum dot lasers

In the previous section the sources of noise were described, and the LEF was introduced - it couples carrier density fluctuations to phase noise. QD lasers have a lower LEF than quantum well (QW) lasers, hence can reach lower linewidths. Another feature of QD lasers is their resilience to external feedback. This section will cover the structure of QDs, and examine the

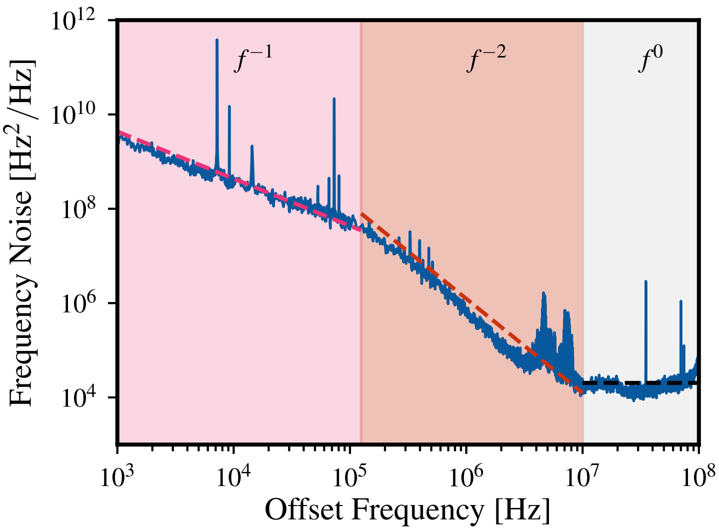

Figure 2.2: The frequency noise of a QD laser plotted against offset frequency. The data used in the plot is taken from [9]. The dependence on frequency is fitted. The pink section shows pink noise (f^{-1}), the red section shows random walk noise (f^{-2}), and the grey section white noise (f^{0}).

phenomena which give rise to these desirable characteristics.

2.1.1 Linewidth enhancement factor

Whilst QW lasers typically have a LEF in the range from 3 - 5 [11], QD lasers offer a near 0 value - values as low as 0.13 have recently been reported [12]. The value of the linewidth can be understood by studying the gain curves for QW and QD lasers. The shape of the gain curve is modulated by the joint density of states and the Fermi-Dirac occupations in the conduction and valence band. QWs have a step-like joint density of states, whereas the 3-D confinement in QDs produces a series of delta function peaks. Variation in QD size leads to inhomogeneous broadening of the energy levels, which is shown in Figure 2.3 (a). The result is the formation of a quasi band of states around each energy level, as shown in Figure 2.3 (b).

The gain of a material is given by the imaginary part of the optical susceptibility, χ, and the real part determines the change in refractive index, Δn; the two are linked by a Hilbert transform [13]. Since the Hilbert transform of an even function is an odd function, $\Delta n = 0$ for the symmetrical QD gain spectrums. For QWs Δn is a finite value. This is illustrated in Figure 2.4.

Practically, manufacturing constraints mean that the size distribution - and therefore the

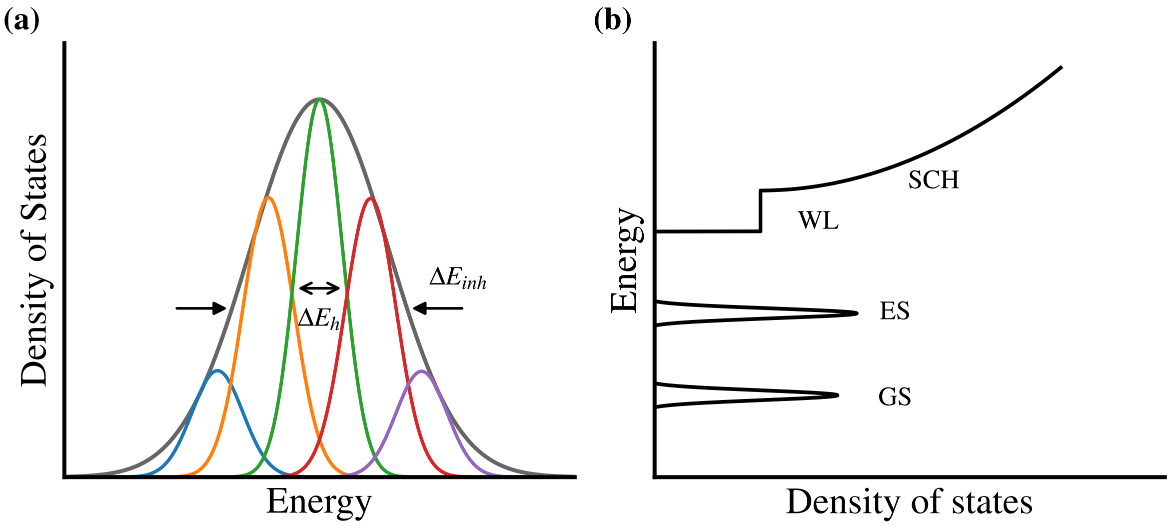

Figure 2.3: (a) A schematic showing the effect of homogenous (ΔE_h) and inhomogeneous broadnening (ΔE_{inh}) in a QD system. (b) A schematic showing the density of states of a QD structure, which includes inhomogeneous broadening. The WL and SCH density of states are also shown.

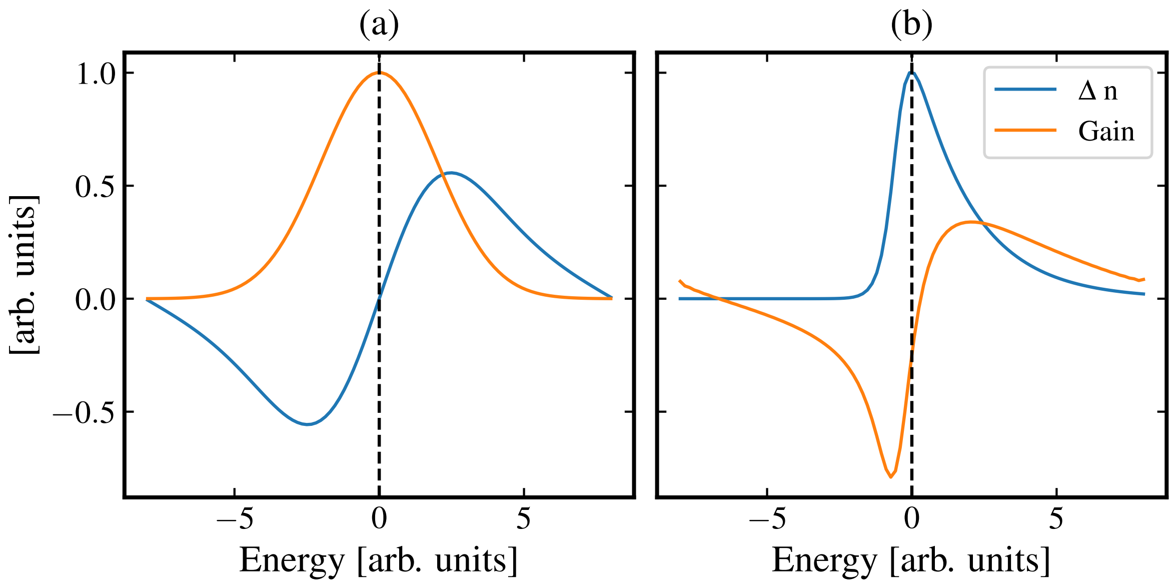

Figure 2.4: The gain spectrum, and its Hilbert transform, which represents the change in refractive index for (a) a QD system where the gain follows a Gaussian distribution and (b) a QW system, where the gain is approximated as a sigmoid multiplied by a Boltzmann distribution.

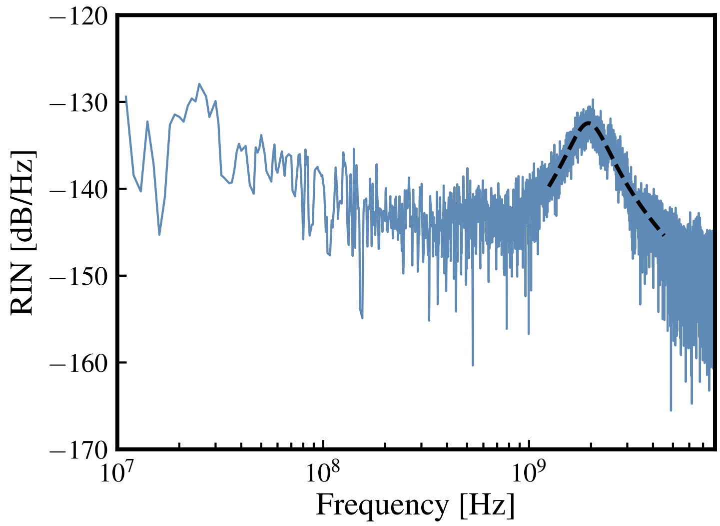

Figure 2.5: The RIN spectrum for a QW device. The key feature is the gain peak, which is fitted with equation 2.6. The data used in this plot is taken from [9].

gain spectrum - is not symmetric. Fluctuations in carrier density alters the refractive index of a material, which is termed the Drude effect [14]. This is not captured by the real part of χ. In QD lasers the LEF depends on current above threshold. Electrons can continue to fill up the excited states, even once the ground state is clamped at the lasing threshold. Consequently, the total electron number increases above threshold, and the Drude contribution to Δn rises. This rise is comparatively small, due to the limited occupancy of the excited states.

2.1.2 Damping factor

Another attractive property of QD lasers is their resilience to feedback due to their high damping factor. This section will investigate what this damping factor is, and how this resilience arises. To explain the damping factor, we introduce the relative intensity noise (RIN) as the PSD of intensity fluctuations. An example RIN spectrum is shown in Figure 2.5. The RIN is ultimately limited by shot noise. A key feature in the RIN spectrum is the peak, which in the example spectrum is centred at ≈ 2 GHz. This peak can be modelled by a Lorentzian [9]:

$$\mathrm{RIN}(\omega) = \frac{a + b\omega^2}{(\omega^2 - \omega_{\mathrm{RO}}^2)^2 + \omega^2\gamma^2}, \tag{2.6}$$

where ω_{RO} denotes the resonant frequency of the relaxation oscillation, ω is the angular frequency, γ is the damping coefficient, and a and b are dimensionless coefficients used for curve-fitting to experimental data.The resonant frequency of the laser cavity can be expressed as [15]:

$$\omega_{RO}^2 \approx \frac{v_g a N_p}{\tau_p}. \tag{2.7}$$

Here v_g is group velocity of light, a the differential gain ($\partial g/\partial N$) of the active region, N_p the average photon density in the cavity, and τ_p is the cavity photon lifetime. Cavity geometry is important in this value, as well as the gain characteristics of the material. The height of the peak is determined by the rate at which injected carriers cascade into the lasing state, replacing carriers that have recombined. This parameter is called the damping factor, γ. Nonradiative recombination shortens the total carrier lifetime and hence significantly increases this damping factor [16]. When light is reflected back into the laser cavity the coherent state may collapse if the optical feedback is strong and its frequency matches the cavity resonance. A high damping factor increases this tolerance to feedback. The K-factor is a parameter which describes how the damping changes as a laser is more strongly driven. The K-factor is expressed as [15]:

$$\gamma = K f_{RO}^2 + \gamma_0, \tag{2.8}$$

where γ_0 is given by $1/\tau_c$, the inverse of the differential carrier lifetime. QD lasers typically have strong damping factors, even when the driving current is small - a typical K value is 5 ns. On the other hand, a typical K value for a QW laser is 0.8 ns [9]. The following section will describe how feedback can be used to reduce laser linewidth, taking advantage of this high QD damping factor.

2.2 Self injection locking

Self injection locking is a technique where coherent optical feedback is provided by an external optical resonator. The external resonator acts as a narrow-bandwidth frequency-selective filter. These narrow-bandwidth photons are fed back, allowing the laser to operate at the injected frequency with reduced noise. The strength of the feedback determines the degree to which linewidth is reduced. The performance can be separated into 4 regimes which depend on the feedback strength [15]. These phases are described below in order of increasing feedback strength.

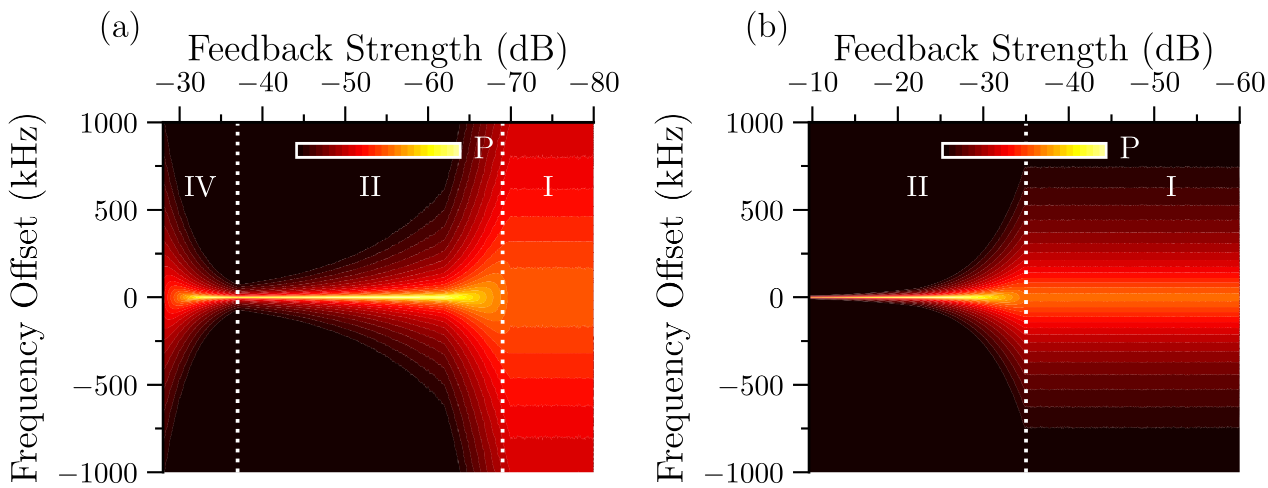

Figure 2.6: Simulated power spectral density as a function of feedback strength for a (a) QW laser and (b) QD laser. The data used in this plot is taken from [9].

1. Regime I: At weak feedback, only one mode of each original internal mode exists. At some feedback phases the laser linewidth is broadened, and at others it narrows.

2. Regime II: Multiple solutions are allowed for each original laser mode. At some feedback phases the laser linewidth is broadened, and at other it narrows.

3. Regime III: The laser linewidth remains narrow regardless of feedback phase. Mode hopping is suppressed. This regime occupies a very small range of feedback powers.

4. Regime IV: Coherence collapse - relaxation oscillations are resonantly driven and mode hopping may take place, resulting in a dramatic broadening of the linewidth.

These regimes are illustrated in Figure 2.6 (a) and (b), which shows the differing response to feedback for a QW and QD laser with an external cavity. The QW enters regime II at ≈ -70 dB, and linewidth reduces steadily until ≈ -37 dB, where coherence collapse occurs, and the linewidth broadens dramatically. For the QD case, even at high feedback strengths, coherence collapse does not occur, owing to the high damping factor. It should be noted that in both regimes Regime III is not identified, as the linewidth does not remain invariant over a feedback range.

Chapter 3

Literature Review

This section will review the structure of QD lasers, and how they are fabricated. This will be followed by reviewing a simulation framework for QD lasers. The experimental performance of these lasers will be discussed, and the effects of optical feedback will be examined.

3.1 Structure of QD DFBs

Distributed feedback lasers (DFBs) are one variety of lasers used to achieve low linewidth. The cavity incorporates a Bragg grating distributed along the waveguide length, such that only certain wavelengths are reflected. This reduces the bandwidth of the reflected light, such that only a single mode propagates in the cavity.

The most common QD gain media consist of InAs QDs grown on an InGaAs layer using solid source molecular beam epitaxy. Historically, these structures were grown on GaAs substrates. For scaleable mass production, integration of lasers on silicon is key. Growth of QDs on a silicon substrate is a relatively new field, complicated by the lattice mismatch between III-V gain materials and silicon. This mismatch causes threading dislocations, which act as non-radiative recombination paths, such that the laser requires higher injection currents to reach threshold. Researchers at UCL have developed a method to produce near-defect free active regions through growth of dislocation filter layers and thermal annealing, which has been used to fabricate QD DFBs on silicon [4]. The cross section of this DFB is shown in Figure 3.1.

Subsequent work on QD DFBs grown on silicon has achieved low free-running linewidths. For example, Dong et al. report a free running linewidth of 41 kHz. This linewidth can be further reduced under EOF. Since the free-running linewidth is below the 200 kHz threshold,

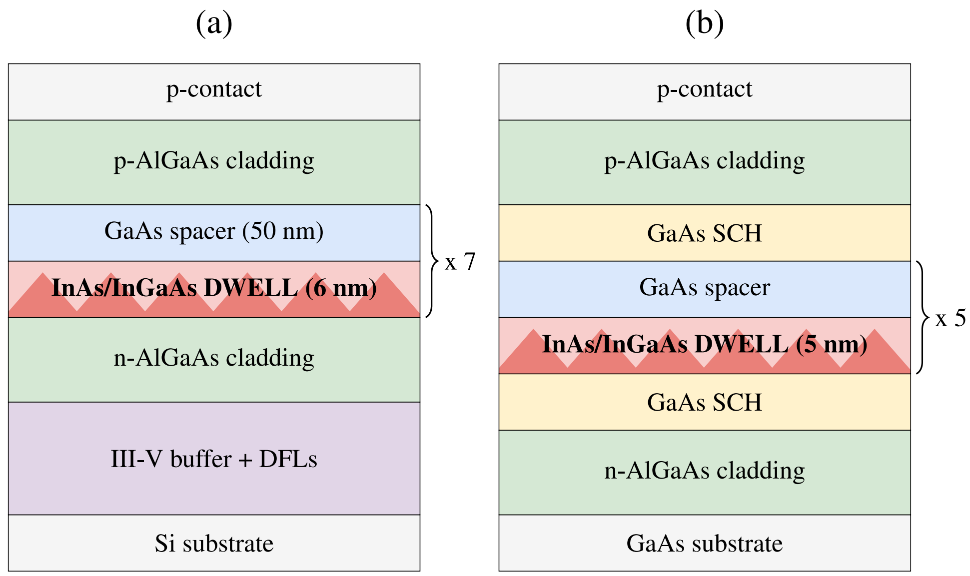

Figure 3.1: Schematics showing the laser structure of InAs/GaAs QD lasers. (a) shows a QD laser grown on Si by [4] (b) shows the structure of the QD DFB simulated by G&R, experimentally tested by [17].

QD DFBs are a viable system for CV-QKD, and such lasers have already been grown on Si using UCL facilities.

3.2 Simulation of QD DFBs

QD lasers are more complicated systems than QW lasers. The key difference is that the populations across all carrier states must be tracked. In addition, the resulting gain and optical properties, as well as the propagating field must be simulated, and these parameters are linked. Gioannini and Rossetti [14] (G&R) present a time-domain-travelling wave model (TDTW), which considers the dynamics of the charge carriers through rate equations. TDTW models simulate the electromagnetic field across the length of a cavity over time. This is tracked by solving the forward and backward travelling wave equations within the cavity.

The system that they model is shown in Figure 3.1 (b). This uses a similar material structure to the QD DFB fabricated at UCL by Wang et al [4]. QD lasers grown on Si still have higher defect densities than those grown on GaAs (10^6 vs 10^4 cm^-2 respectively), which impacts the laser dynamics. For example, QDs on Si have a higher recombination rates, which increases the damping factor and tolerance to optical feedback. The G&R model was identified as a strong foundation, because it contains flexible parameters, that could easily be adapted to simulate different QD DFB structures. Across QD lasers, Gioannini and Rossetti (G&R) identify 4 key factors to include:

1. Homogeneous and inhomogeneous broadening of the gain spectrum. This determines the gain spectrum shape and spontaneous emission characteristics.

2. Electrons and holes have different dynamics. Owing to the increased effective mass of holes, the energy level separation is smaller between hole states.

3. The refractive index fluctuates with the carrier density - the Drude effect.

4. Longitudinal spatial hole burning caused by non-uniform photon density. The carriers are almost all confined within the QDs, hence carrier diffusion is strongly suppressed (especially compared to a QW laser).

To address the first key factor in modelling QDs, the QD population is divided into 51 groups. The transition energy of these groups assumes a Gaussian distribution around the central ground state energy. The standard deviation is chosen as 38 meV, which represents the inhomogeneous linewidth broadening. The carrier populations in each of these states are tracked through individual rate equation, and a spontaneous emission term is defined for each state.

The structure of these rate equations captures the second key factor - differing behaviour of electrons and holes. The structure of the energy levels of an InAs/GaAs QD is shown in Figure 3.2 (a). The energy spacing between hole states is much less than that of electrons states, such that thermal excitation redistributes holes. Consequently, the QD hole population remains in quasi-thermal equilibrium - it always follows a Fermi-Dirac distribution. This is not the case for the electrons. The structure of the rate equations for electrons is summarised in Figure 3.2 (b). Transitions between the states are modulated by escape and capture times (τ_e and τ_c) and the population of the state the electron is in. A key feature is the Pauli Blocking factors, which describe the occupation of the state an electron is moving to. For example the rate at which electrons are captured by the ground state (GS) from the first excited state (ES1) depends on the number of electrons in ES1, the capture time constant, and the occupation of GS (how much room is left in the GS). It should be noted that Pauli blocking factors are not included in equations involving transitions to the wetting layer (WL) or separate confinement heterostructure layer (SCH), because these levels have a continuum of states. The full system of the equations can be found in the reference, but two corrections should be made, as described in Appendix 7.2.

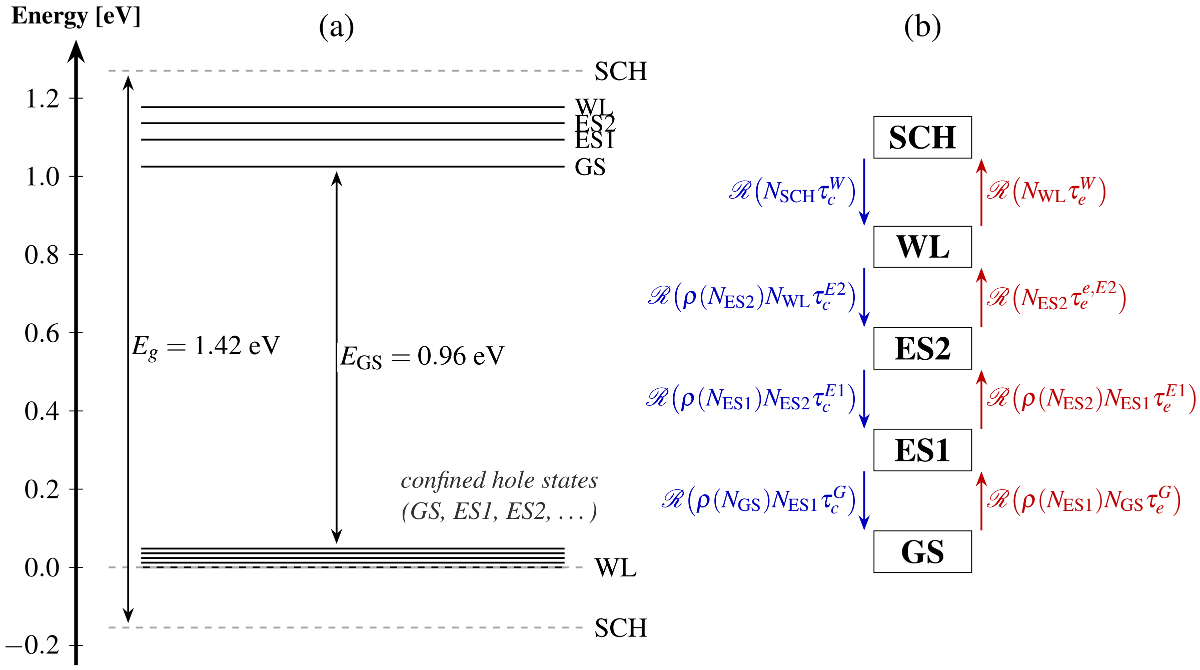

Figure 3.2: (a) Energy level diagram of an InAs/GaAs QD. (b) Schematic showing the variables which determine the electron capture and escape rates. ρ represents an occupation probability used to calculate the Pauli blocking factor.

The final key factor is addressed by discretising the waveguide in 1-D, such that rate equations are solved for a single segment, which impacts the state of the neighbours. Since lateral carrier diffusion is not included, the authors assumed an effective width larger than the experimental design they measure (2.2 μm vs. 1.4 μm) to account for carriers injected through the ridge, which diffuse in the lateral direction under the ridge. The authors also assume that capture and relaxation times are constant across carrier densities. Ultimately, the relaxation and capture times in the model are fitted parameters, which means that they do not directly inform quantifiable design decisions. Although the complicated physics in a device is not completely captured, the approach provides a first principles basic understanding of device properties. Importantly, the authors find good agreement between the gain spectrum, L-I curves, and modulation response compared to experimental results in the same system [17].

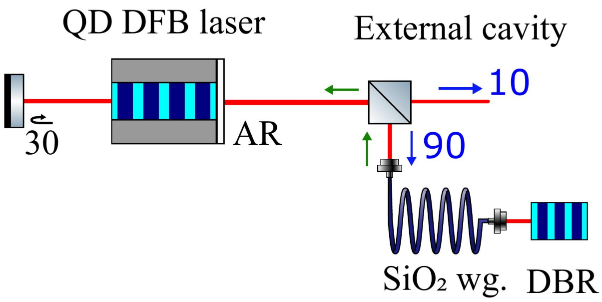

Figure 3.3: Schematic showing a setup for external optical feedback which uses a silica waveguide terminated by a distributed Bragg reflector (DBR).

3.2.1 Experimental locking circuits

The following section will describe how external optical feedback is used to reduce the linewidth in QD lasers. Dong et al. present a locking circuit for a QD DFB which consists of a 10 m optical fibre cavity with a distributed Bragg reflector (DBR) at the end. This is shown in Figure 3.3. This design achieved a feedback tolerance of -8 dB, and a linewidth of 0.6 Hz.

To reach low linewidths, the grating parameters of the DBR must be tuned carefully. The noise reduction factor (NRF) determines the locked linewidth according to $\Delta f_{locked} = \Delta f_{free}/NRF$ (for a system where thermorefractive noise is not high) [18]. The NRF below coherence collapse is expressed as:

$$\mathrm{NRF} = [1 + KL_{\mathrm{ext}} \tanh(\kappa L)]^2, \tag{3.1}$$

where K is the effective optical feedback coupling factor, L_{ext} is the external cavity length, κ is the grating strength, and L is the resonator length. The product κL is a dimensionless parameter referred to as the coupling strength, which describes the strength of the DBR reflection.

As the coupling strength increases, the reflection increases, and the Q factor decreases. The Q factor of the DBR is defined as the central wavelength of the resonance peak divided by the reflection bandwidth. High Q DBRs produce a narrow width spectrum, but the feedback strength is weak. This is depicted in Figure 2.6 (a).

Ultimately there is a trade-off between feedback strength and bandwidth. The range of coupling factors can be divided into two regimes: one where the feedback strength limits the linewidth, and one where the bandwidth is too broad, triggering coherence collapse. This is illustrated in 3.4 (b).

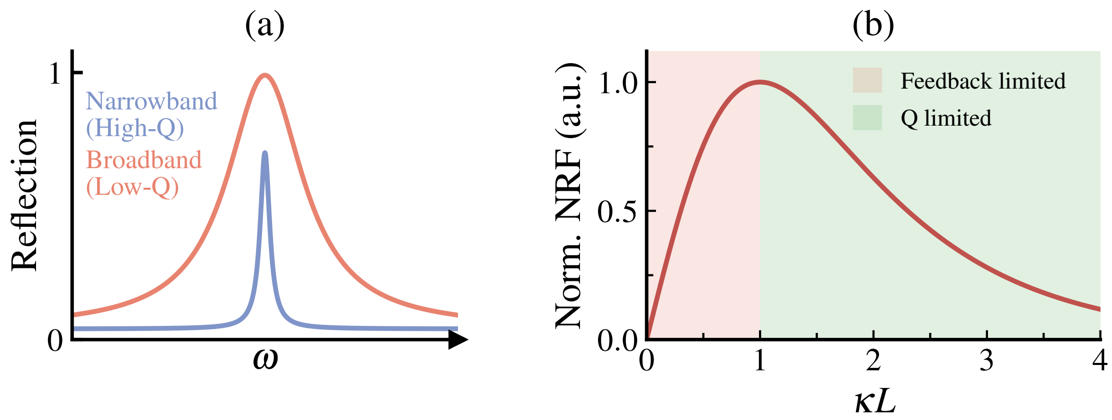

Figure 3.4: (a) An illustration of the trade-off between bandwidth and reflectivity of low and high Q resonators. (b) NRF plotted against κL showing two regimes. In the red regime, the Q factor is high, but the feedback strength is weak. In the green regime the feedback strength is high, but the Q factor is too low.

Chapter 4

Methodology

This section will describe the method to simulate the optical properties of an active QD material. We will then examine how these properties can be used to simulate a DFB system. PICWave will be introduced as a software to perform TDTW, with scope to incorporate coupled physical effects. We will then explain how the dynamics of the free-running laser were simulated, followed by the dynamics under EOF.

4.0.1 Details of PICWave TDTW model

PICWave is a commercial software used to model the interaction of multiple optical components in a photonic circuit. It includes an active model, such that the optical properties of a gain material can be specified, and the laser diode can be incorporated in a photonic circuit. The PICWave optical model uses a 1D TDTW solver. A circuit is comprised of sections, each of which is split up into elements in the z-direction. The timestep is chosen by the amount of time it takes for a photon to travel a single z element. PICWave uses the slowly-varying envelope approximation - the optical field is divided into a rapidly oscillating factor at a central frequency, and a complex envelope which slowly varies relative to the optical period. The fast term gives rise to constant phase factors, so only the time evolution of the envelope must be tracked. A similar approximation is made by G&R. The PICWave carrier model splits the active region into elements along the z direction. In each z-slice a standard rate equation is used to determine the carrier density at each timestep [19].

4.1 Simulation of material properties

PICWave allows for simulations involving multiple carrier levels, although there are limitations. Firstly, to model the gain associated with a transition, a gain lookup table must be imported. The gain table is a function of carrier density of the upper level, N, and wavelength, λ. Generating this gain table requires simulation outside of PICWave. This was performed in Python, by modelling the rate equations detailed by G&R. The list of parameters included in the simulation which are not specified in G&R are included in Appendix 7.3. The stimulated emission term was neglected, to prevent gain clamping above threshold, such that only the material gain is calculated. The differential equations were solved using a first order Euler method, until the number of electrons and holes had converged for all energy levels at a given current. A gain table was produced by performing this step over a range of input currents from 0 mA to 1728 mA. Separate gain curves were saved for the GS and ES1 transitions to be imported into PICWave. The spontaneous emission rates were also calculated and imported into PICWave. The parameters m_h^*, m_e^* and the energy levels between transitions (which are not detailed in the paper) were tuned to match the curves presented in the paper, which match experimental values.

The second challenge of importing an optical model to PICWave is that variables may not be shared between multiple levels. In the G&R model, for example the capture rate from ES1 to GS depends on both populations. In PICWave, the rate can only be specified as a function of the ES1 population. This presents problems in calculating Pauli blocking factors. To circumvent this, the occupation probabilities of each level were calculated as functions of a different carrier population. For example $\rho_{GS}(N_{GS})$ was approximated as $\rho_{GS}(N_{ES1})$ using steady state population data from the Python simulation. This captures the effect of the Pauli blocking factors, as well as information about the hole distribution. An example of the calculated Pauli blocking factors is shown in Figure 4.1, which were fitted with the function $y = 1 - (1 - x)^{\alpha}$.

4.2 Simulation of QD DFB

The gain and spontaneous emission spectra calculated from the previous step were imported into PICWave using the wide band model importer. This imports material gain and spontaneous emission rate against wavelength as a function of carrier density and temperature. The gain spectra are fitted over a range of 0.031 μm around the central gain peak, which occurs at 1.29

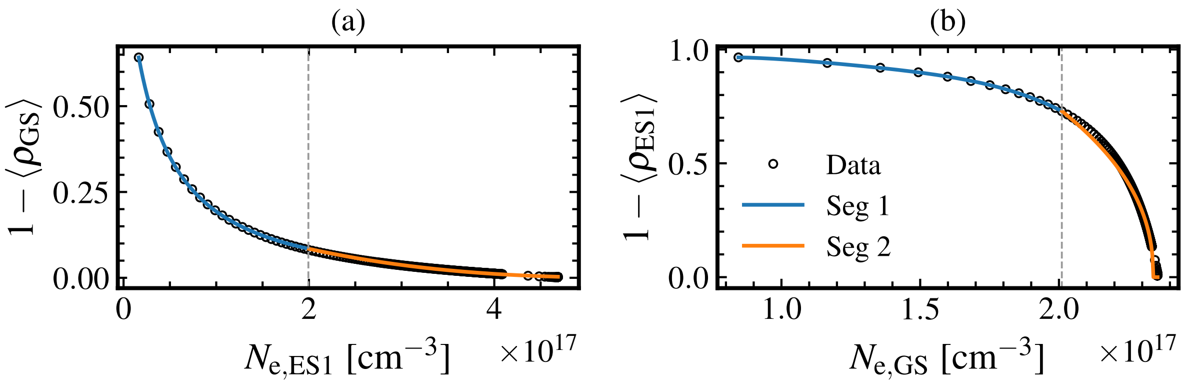

Figure 4.1: Pauli blocking factors as a function of a neighbours occupation (a) ES1 $\rightarrow$ GS (b) GS $\rightarrow$ ES1

μm. As established in the previous section, the temperature was fixed at 25 °C, and thermal phenomena were neglected. The rate equations, which include the Pauli-blocking factors, were encoded in the dynamics of the active material. The L-I curve, and population levels were modelled using the standard procedure detailed in the PICWave manual. A "zstep" of 6 μm was used. The RIN and frequency noise spectra were calculated as described in the PICWave manual [20] with a total simulation time of 12000 ns, allowing 50 ns for carrier level stabilisation. The linewidth was calculated using the white noise region of the frequency noise spectrum.

To benchmark the performance of the cascade model, a single level model was also incorporated to PICWave. This uses the same gain model as the multilevel model. However, only a single lasing state is considered, whose recombination parameters are calculated as a function of the total carrier level.

4.2.1 Parameter optimisation

With the time constants specified in the G&R paper the simulation would crash, or fail to reach a lasing state. Therefore the time constants and Pauli blocking factor shapes were treated as free parameters to be optimised. Two metrics were used to judge the performance of a parameter set - the L-I characteristics and the population characteristics. The parameter sets were optimised against two cost functions. The first was the root mean squared error (RMSE) of the L-I curve in comparison the G&R result. The second was the RMSE compared to the G&R population densities at 50 mA.

The optimisation process was performed in three stages. First, the parameter space was

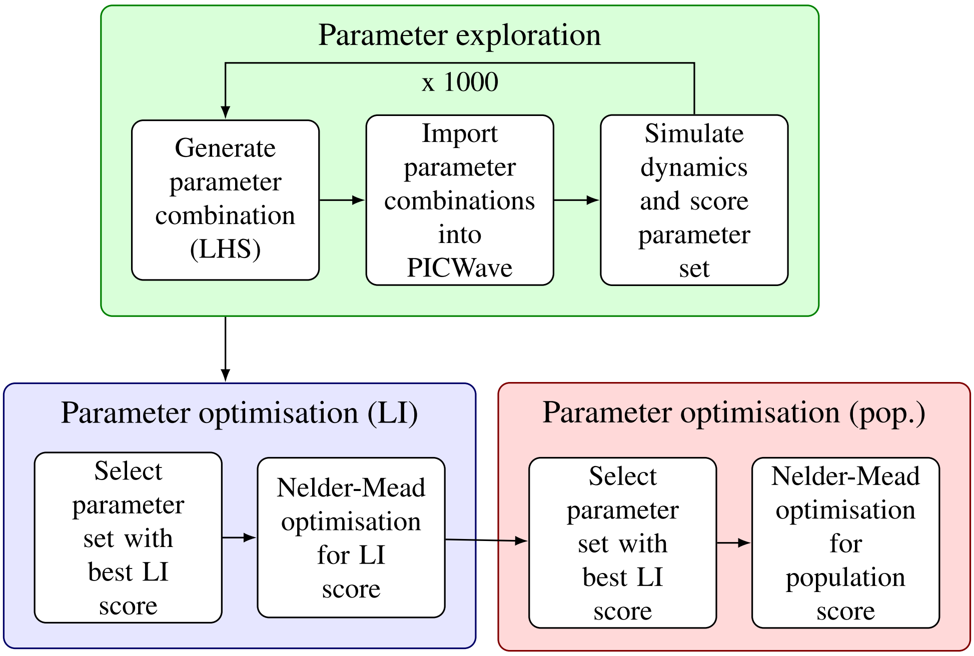

Figure 4.2: A flowchart showing the three stages used to find the optimal parameter set.

explored to identify an approximate parameter combination which finds the best L-I score. The parameters were normalised and sampled in logarithmic space over a range from 0.1 τ to 10 τ. The Pauli-blocking factors were similarly scaled by a stretch factor, $\rho_{stretch}$, over the same range. The sampling method used was Latin Hypercube Sampling (LHS). LHS ensures that parameter space is evenly explored [21] - however per-parameter bounds were narrowed adaptively after invalid runs to prevent crashes, reducing the overall sampling space. This method was used to generate 1000 different combinations of parameters in 20 dimensions.

These parameters were tweaked in a smaller range using the Nelder-Mead method - a numerical method to search for optima in functions - to optimise the Li score over 1000 iterations. A third Nelder-Mead optimisation step was then performed, changing only the time constants. This time, samples were scored against their population score. A summary of this method is shown in Figure 4.2.

The LHS parameter combinations and score data were also used to train a surrogate Gaussian Process (GP). A GP is a supervised learning method, which can be used for regression. A GP treats outputs as a multivariate-normal distribution. The covariance between these outputs is called the "kernel". For this application the Matérn kernel was chosen for its flexibility, and a white noise kernel was added [22]. To understand why a GP finds a parameter important, a SHAP test (Shapley Additive Explanations) can be used. This quantifies the contribution of each input variable to the model's predictions. Positive SHAP values increase the score (a better fit), and negative SHAP values decrease it. The SHAP scores will be considered in combination with the physical processes to understand how different parameters affect the model.

4.3 Simulation of QD DFB under EOF

An external cavity was added to the circuit. The circuit design is shown in Figure 3.3. The circuit connects the DFB cavity to a 10 cm long waveguide, with a 3 mm DBR at the end. The refractive index of both the waveguide and DBR was set to 1.3, which is lower than that of an optical fibre. This was an oversight in the method, and has the effect of reducing the noise coupling factor, K.

The coupling parameter of the DBR was adjusted to change the feedback strength. Dong et al. [18] suggest that coherence collapses at $\kappa L \approx 1$. This was used to inform the simulation range, across which linewidth was monitored. This was followed by measuring the linewidth at different currents, as a function of feedback strength. The linewidth was measured over a period of 200 ns, where the first 20 ns were discarded to allow the electron populations to stabilise.

Chapter 5

Results and Discussion

To begin, the optical properties of the QD material will be presented, and compared to the G&R model [14]. Following this the dynamics of the simulated QD DFB laser will be shown. To evaluate the success of the model, we will compare the behaviour against experimental results. We will also compare the behaviour of the single and multi level QD models. Finally, the dynamics under external optical feedback (EOF) will be explored.

5.1 QD material properties

The first step in modelling QD lasers is to simulate the optical material properties. For this section, the main objective is to implement the model produced by G&R, such that good agreement is reached. The differences between the models show the extent to which different model parameters affect the resulting properties.

The modal gain spectrum of the QD material is plotted in Figure 5.1, comparing the values found by G&R and this work. The peak centred at 1190 nm corresponds to the ES1 transition, and the peak at 1294 nm corresponds to the GS lasing transition. The G&R GS gain peak agrees well with our model. The peak positions match to within 0.5 nm, and show the same FWHM to 12 % (this work: 60 nm, G&R: 52 nm). This is partly due to parameter choice, as the heights of the peaks were scaled by a constant factor - representing the unspecified confinement factor in the paper - to match G&R's values. It is important that the shape of the GS peaks are a good match, which indicates that the inhomogeneous broadening effect is captured.

The ES1 fit is, however, less accurate. The ES1 peaks calculated in this work are shifted to smaller wavelengths by an average of 4 nm (this work: 1190 nm, G&R: 1194), but have a

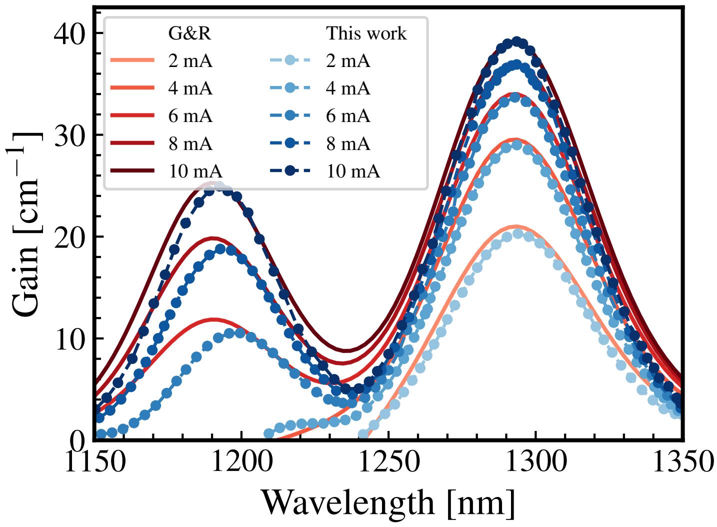

Figure 5.1: Comparison between the gain curves obtained by Gioannini and Rosetti [14], versus those calculated in this work using similar parameters.

similar FWHM compared to G&R's values: 39 nm and 38 nm respectively. This is attributed to unspecified parameters which are used in the G&R method - importantly, the energy spacing between ES1 and ES2 peaks and the electron/hole effective masses. The precise position and shape of the ES1 peak is not key in modelling this system, as the DFB grating is designed for GS lasing. It is important, however, that the tail of the ES1 peak coincides with the GS peak, as this breaks the GS peak symmetry, such that the α factor at the peak is non-zero. This condition is fulfilled.

5.2 Static and dynamic laser characteristics

5.2.1 Parameter optimisation

The $\rho_{stretch}$ parameters applied to the Pauli blocking factors are 1.276 for the GS $\leftrightarrow$ ES1 transitions, 4.3 for the ES1 $\leftrightarrow$ ES2 transitions, and 6.6 for the ES2 $\leftrightarrow$ WL transitions. Their effect on the Pauli blocking factors is demonstrated in Figure 5.2. Scaling was necessary to ensure that the simulations were numerically stable. This stretching has significant implications on the capture rates, especially the $ES1 \rightarrow ES2$ blocking factor. The ES1 population is always below 1×10^{18} cm^{-3}, so this cuts off escape from ES1 to ES2. This is observed to a lesser extent for the GS $\rightarrow$ ES1 transition.

The fitted time constants are shown in Table 5.1. Generally the escape and capture time

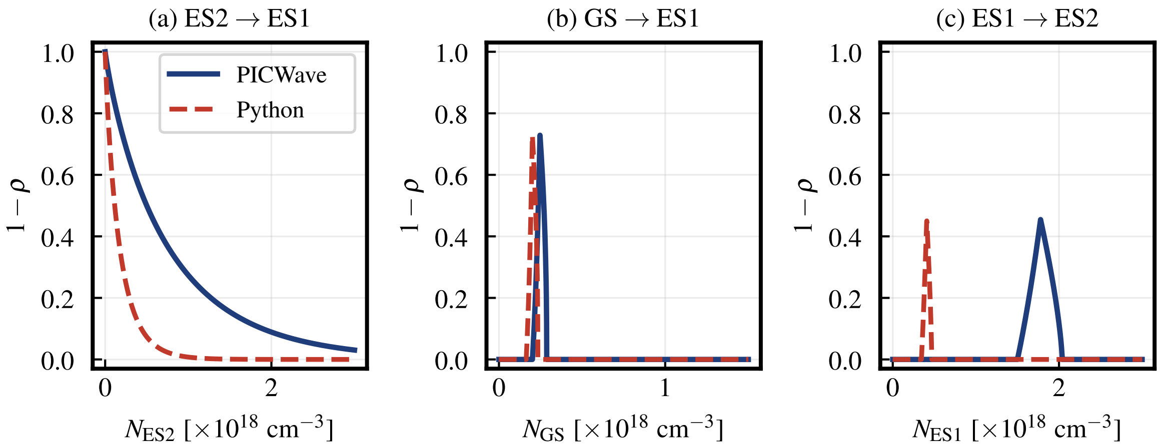

Figure 5.2: The Pauli blocking factors are compared between the Python model, and the stretched PICWave models for 3 different transitions.

constants are found to be larger in this work by a factor of ≈ 10 when compared to G&R. The Auger recombination time constants are similar between the two models, but the spontaneous emission time constant are on average 13 times smaller in the PICWave model compared to G&R.

Experimental four wave mixing measurements of the radiative lifetime suggest a radiative GS recombination time of 633 ps [23]. This is in fact closer to the Nelder-Mead optimised solution than the value suggested by G&R. On the other hand, the GS rise time (WL→ES2→ES1→GS) is experimentally measured at 2.6 ps. The rise time found by the Nelder Mead optimised parameters is much larger than this. As a result of scaling applied to the Pauli-blocking factors, the capture times are not directly comparable to physical values. Although their relative magnitudes still yield useful information about the carrier dynamics, they correct for secondary effects. Notably, inaccuracies in the Pauli Blocking factors. Despite this the model's population dynamics are similar to G&R's model, which will be the focus of the following section.

5.2.2 Population dynamics

Firstly, we will cover the population of the energy levels. In a physical system these parameters are not directly measurable, but they are crucial in determining the L-I characteristics and linewidth.

Figure 5.3 (a) shows the evolution of the population states close to the threshold current

Table 5.1: Comparison of QD-DFB time constants: Gioannini and Rossetti versus the optimised parameter set in this work.

Parameter	G&R (2011) [ps]	RO model [ps]	Ratio
$\tau_c^{e,W}$	1.20	1.19	0.99
$\tau_e^{e,W}$	2.87	2.85	0.99
$\tau_c^{h,W}$	23.20	–	–
$\tau_e^{h,QD}$	469.60	–	–
$\tau_c^{e,ES2}$	3.00	28.08	9.36
$\tau_e^{e,ES2}$	59.11	552.41	9.35
$\tau_c^{e,ES1}$	2.00	20.17	10.08
$\tau_e^{e,ES1}$	7.29	49.19	6.74
$\tau_c^{e,GS}$	2.00	10.36	5.18
$\tau_e^{e,GS}$	15.51	80.35	5.18
τ_{Sp}^{GS}	2800.00	276.90	0.10
τ_{Sp}^{ES1}	2800.00	189.86	0.07
τ_{Sp}^{ES2}	2800.00	149.99	0.05
τ_{Au}^{GS}	660.00	1157.80	1.75
τ_{Au}^{ES1}	275.00	710.12	2.58
τ_{Au}^{ES2}	110.00	146.79	1.33
$\tau_r^{e,W}$	100.00	1328.40	13.28

over a period of 15 ns. The Python model is compared to the PICWave model. Across both models, the populations evolve over a similar timescale, flattening by 2.5 ns. The PICWave populations evolve less smoothly than the Python model - there are kinks in the evolution spectrum - for example a dip is observed in the ground state population at ≈ 1 ns. This is attributed to the high spontaneous emission rate, which is increased by a factor of 10 compared to G&R. Consequently, when a carrier level exceeds its maximum occupation, it is quickly depleted, causing a sharp drop. The carrier density between final states differs by an average of 26 %, but the qualitative trend in carrier density is the same.

The occupation probabilities of the QD states are shown in Figure 5.3 (b) for the Python and PICWave models. A key feature of QD carrier dynamics is the clamped ground state, and unclamped excited states. Although, ρ_{ES1} and ρ_{ES2} show discontinuities at the threshold current, this behaviour is qualitatively captured. The ES1 and ES2 populations are not clamped above threshold, but differ by 0.1 for ES1 and 0.2 for ES2 when compared to G&R's model.

The PICWave GS electron occupation probability clamps at at ≈ 0.82 at I_{th} (5.74 mA),

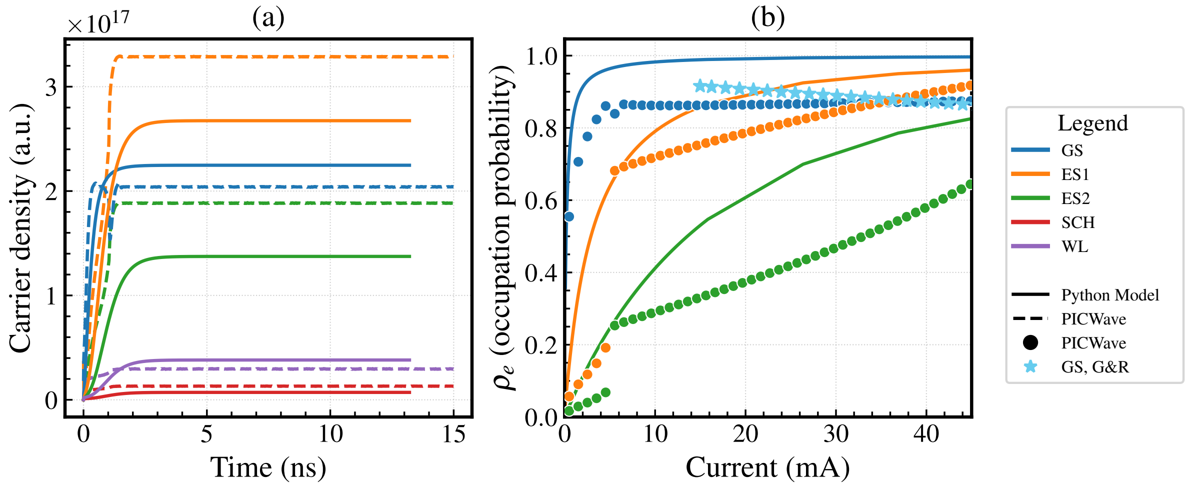

Figure 5.3: (a) Carrier density in quantum dot states against time for a current close to threshold - 7 mA (b) Occupation probability for quantum dot states as a function of input current.

whereas the G&R value decreases with increasing current. This reveals a flaw of the PICWave model. It does not directly model the hole populations separately. Although the gain is clamped, it is a function of ρ_e^{GS} and ρ_h^{GS}. The decrease in electron probability can be balanced by an increase in hole occupation. The PICWave model only considers the electron density. To compensate, the PICWave model calculates gain based on 89% of the GS electron population. This also improved simulation stability. In the python model, stimulated emission was not included, hence clamping is not observed, so ρ_e continues to rise.

To probe parameter effects on population dynamics further, a GP surrogate model was trained on the LHS parameter set and its scores, achieving 10% relative error on 91 test points (R^2=0.53). The model is able to capture the general trend. A SHAP test identifies the capture times from the ES1 and GS as important parameters, as shown in Figure 5.4. This is intuitive, as these parameters directly impact the cascade dynamics between QD states. Spontaneous emission times are less important in the final occupations. This suggests that there is a ridge in the cost function made up of parameter sets with different values. The Nelder-Mead optimisation method may have converged on one of the solutions on this ridge, but this may not be the most physically accurate solution.

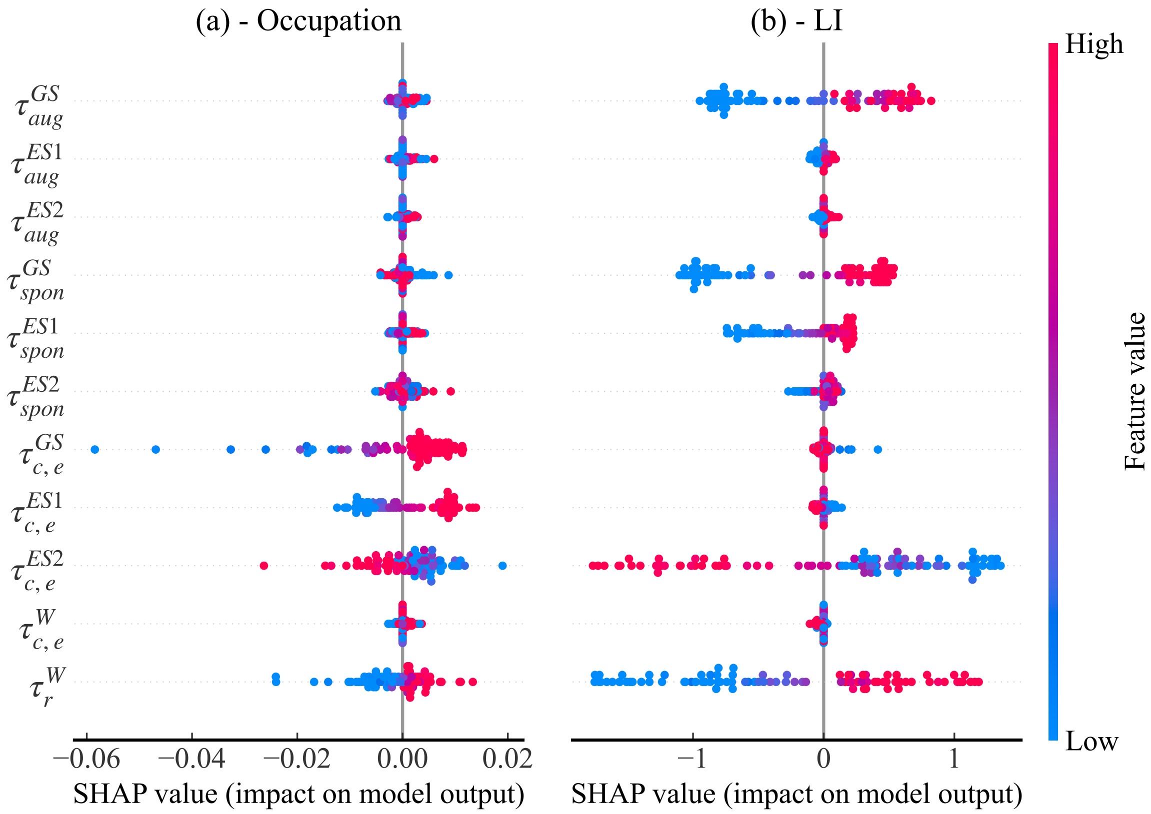

Figure 5.4: ”Beeswarm” plot, which presents the SHAP values across parameters. This shows the relative influence of each parameter on (a) the occupation score and (b) the L-I score. The SHAP value is not normalised, but proportional to the size of the score. Since the RMSE is smaller for the populations, the SHAP values are much less.

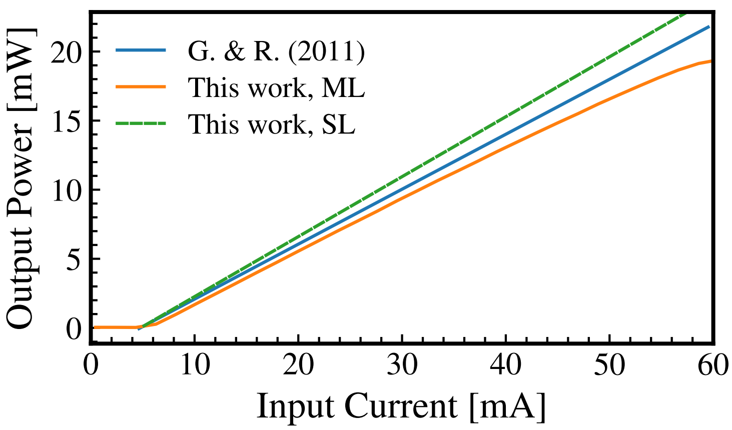

Figure 5.5: Output power against input current comparing G&R [14], and the multilevel (ML) and single level (SL) models from this work.

5.2.3 L-I characteristics

This section will use findings from the population dynamics to explain measurable laser characteristics, starting with the power-current relation (L-I relation). The L-I curve from this work is shown in Figure 5.5, and compared to that calculated by G&R. The single-level is also shown, which successfully reproduces the linear L-I characteristics of the G&R model. Its limitations are more apparent in the dynamic regime, and discussion of this model is reserved for later sections.

The models find similar threshold currents (multilevel model: 5.7 mA, G& R: 4.7 mA). However, the multilevel model L-I curve is not linear, and diverges from the G&R relation as the input current increases (> 30) mA. We will attempt to explain the shape of the LI curve by inspecting the fitting parameters of the model.

The GP accurately predicts the L-I scores of the 91 test data (R^2 value=0.968). Inspecting SHAP values provides insight into the influence of time constants on the L-I fit (Figure 5.4 (b)). This suggests that the GS Auger recombination time is an influential parameter here. Physically, the Auger recombination rate determines how quickly carriers drain from the lasing state, hence has a strong influence on the threshold current. Increasing τ_{Aug}^{GS} would increase the GS population decreasing the threshold current, improving the score.

It is also notable that the τ_c^{ES2} is the most important capture time. For the other capture times, the escape times vary with them as a result of the detailed balance condition. For this transition, however, the escape time is essentially infinite, as a result of the scaled Pauli blocking factor. Therefore this variable can have a large effect on the LI curve, by controlling the rate at which electrons are captured by ES1, without changing the escape time from ES1. A faster cascade rate corresponds with a lower threshold current.

A key difference between the L-I characteristics is the decreasing gradient above 30 mA in this model. Crucially, this is not inherent with the fitted Pauli blocking factors - other combination of parameters produce straight fits, but generally have low ES2 occupancies. The main difference between these ”linear” fits are lower capture times between the QD states. At higher currents, where recombination is higher, a slow rate of electron exchange acts as a bottleneck, such that electrons non radiatively recombine before reaching the lasing state. This is referred to as spectral hole burning, which causes gain compression. Our set of parameters overestimates

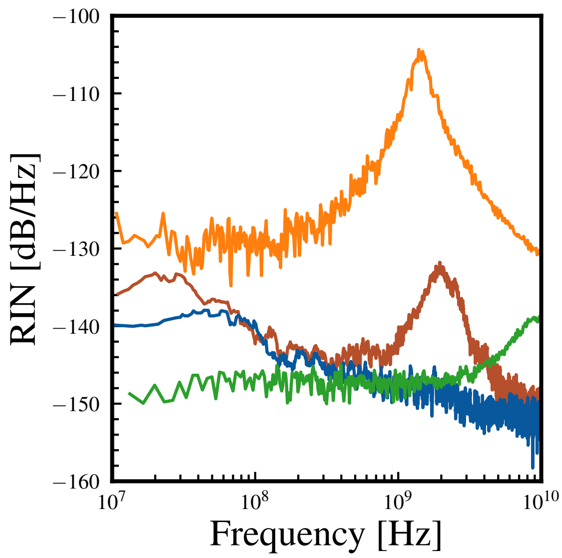

Figure 5.6: Relative intensity noise comparing data from a free-running quantum well laser with free-running quantum dot lasers measured at 0 dBm output power. Experimental data from [9].

this effect.

5.2.4 Relative Intensity Noise

To evaluate the multi-level model, it is helpful to compare the calculated system behaviour to that of the single-level model. As noted previously, the static L-I characteristics are a good fit to G&R. However, as shown in Figure 5.9 there are large differences in the relative intensity noise spectra.

Firstly, the peak in the relative intensity noise spectrum is much larger than the peak of the multi-level model, and of the experimental QD laser. This result is attributed to its under-estimated damping factor in the single level model. The higher energy levels act as a reservoir, damping the effect of injection current shot noise, reducing RIN. The height of the RIN peak is determined by the resonant frequency of the electron frequency, and the damping as electrons are exchanged.

Although the peak in the multi-level model is smaller, it is not completely suppressed, as the experimental QD laser results find. The experimental QW laser and single-level QD models show a peak centred on $\approx 2 \times 10^9$ Hz, whereas the multi-level QD peak is centred at $\approx 1 \times 10^{10}$. The damping factor of the multi-level model was further investigated. The RIN spectrum was

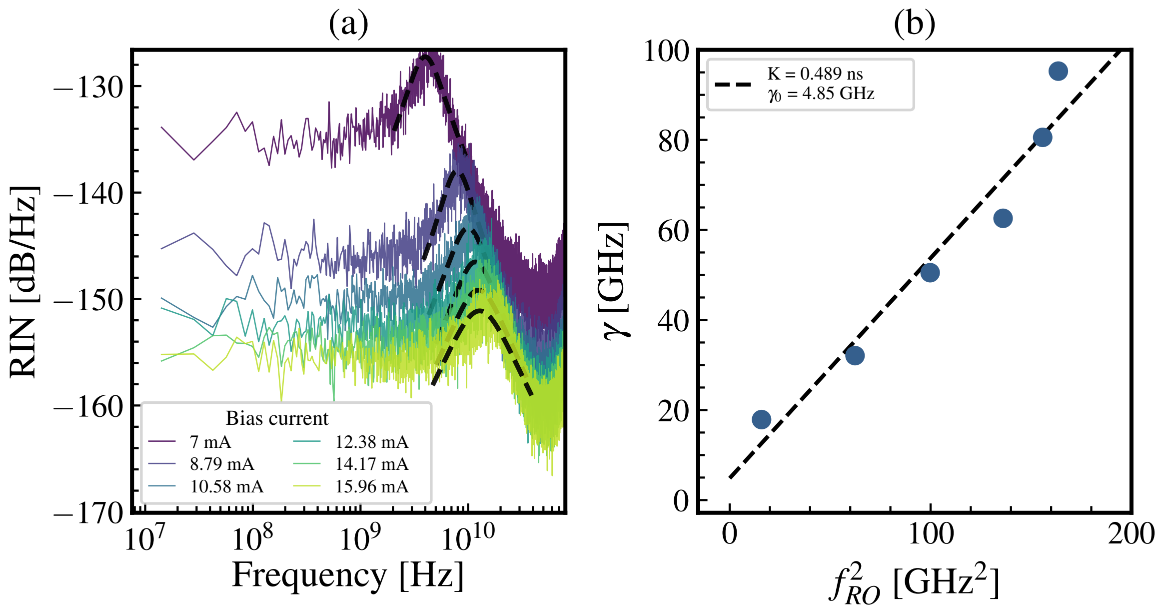

Figure 5.7: (a) Plot of RIN spectrum with peak fitted with equation 2.6 (b) Plot of γ^2 against f_{RO}^2. The K value given by the slope of the best fit line was 0.489 ns, and the γ_0 value, the y intercept, was 4.85 GHz.

plotted for multiple injection currents above threshold, and the plots are shown in Figure 5.7 (a). The peaks were fitted with equation 2.6, and the resulting γ^2 and f_{RO}^2 are plotted in Figure 5.7 (b). The K value given by the slope of the best fit line was 0.489 ns, and the γ_0 value, the y intercept, was 4.85 GHz.

The damping factors, γ_0 are, as expected, a factor of 2 higher than those in QW lasers. However, Dong et al. find that the K value of their QD laser is over 5 ns (which measures how the damping factor scales when the oscillator is more strongly driven), and note that the value for a QW laser is typically ≈ 0.8 ns. Our K factor is therefore a factor of 10 below the expected value. The key reason for this appears to be the resonant frequency of the peaks. The resonant frequency, according to Equation 2.7, depends on the differential gain, and the cavity resonance. An increased differential gain could therefore be the cause of this peak, and this should be investigated with the amplitude modulation technique proposed in G&R [7]. A physical reason for an increase in differential gain could not be suggested with the current results available.This does not seem to be an inherent flaw of the model, as the RIN peak is completely suppressed with different parameter combinations. Overall, the high raw damping factor is encouraging, as it suggests that the model should tolerate high levels of optical feedback.

5.2.5 Linewidth enhancement factor

The linewidth is a function of the spontaneous emission rate and the LEF. To understand the relative contribution of these terms, it is therefore useful to first evaluate the LEF. The α_H factors at threshold are shown for the Python and PICWave models, and compared to literature values. The results are presented in Table 5.8 (a). The α parameters fall within the range found in other quantum dot systems. The α_H value is a material parameter, and is not well defined above threshold. Therefore the effective LEF parameter was also calculated.

Author	I_{th} [mA]	α	method
This work (Py)	5.7	0.87	sim.
This work (PW)	5.7	0.34	sim.
Martinez (2025) [24]	12	0.67	exp.
Gioannini (2006) [7]	3.8	0.44	sim.
Zhao (2021) [25]	-	0.78	sim.
Dong (2025) [18]	12	2.3	sim.

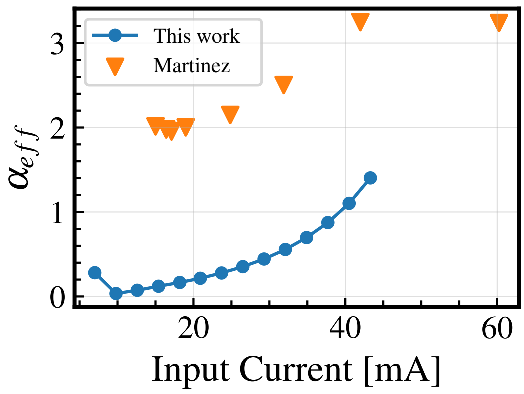

Figure 5.8: (a) Linewidth enhancement factor, α_H at threshold current for quantum dot lasers. (b) High frequency limit of the effective linewidth enhancement factor α_H calculated from the AM/FM response as a function of injection current. Experimental data is from [24].

The α_{eff} are plotted against input current in Figure 5.8 (b). The numerical values from this work are found to be similar to Gioannini et al. 2006, who uses a similar modelling framework. The effective alpha parameter follows the same trend as that established experimentally by Martinez et al. but the values are lower. The change in refractive index in PICWave is calculated using the total carrier density, rather than weighting different states contributions separately. Another factor is spatial inhomogeneity. Although the model in this work calculates the gain spectrum by considering an inhomogeneous quantum dot size distribution, this gain distribution is applied uniformly across the DFB. Physically this is not realistic, and different sections of the gain spectrum would vary. This effect is unquantified, but could be investigated in future work. These calculated values of LEF will be useful in the analysis of the calculated linewidths, as discussed in the following section.

5.2.6 Frequency noise

The simulation of frequency noise is key in understanding frequency drift of the laser, and therefore it's suitability for QKD. Frequency noise data cannot be collected at frequencies lower than 1 MHz in the current model. The resolution is determined by the simulation length, which is capped by PICWave. This means that the flicker noise region cannot be reached. However, there is no sign of increase in frequency noise at frequencies lower than 10^6 Hz. Thermal effects are not considered in this model, hence the thermo-optic fluctuations are not present. Additionally, there are no additional environmental factors which modulate the current, although this could be included in future simulations.

In the white noise region of frequency noise, $10^7 - 10^8$ Hz, the multi-level QD model shows an average frequency noise of 5.0×10^4 Hz^2/Hz. The calculated linewidth is therefore 160 kHz. Again, the single level QD model shows a higher average frequency noise of 5.2×10^6 Hz^2/Hz, and therefore a linewidth of 16 MHz. The literature linewidth of a free running QD laser - calculated from the white noise level - is 61 kHz.

The single level model has a much higher linewidth than the experimental values. We suggest that the rate of spontaneous emission in the single level model is overestimated. In the multiple states model the emission rate from each level is considered individually. The cascade dynamics and individual state emission rates appear to have a large effect on this value.

The linewidth of the multiple-level model is more similar to the experimental value. In fact, it falls in the range of other free-running QD DFB lasers. For example Wang et al found a linewidth of 243 kHz [26], and Wang et al. a linewidth of 157 kHz [27]. A key consideration when discussing the practicality of this model is its response to transient changes. The rate equations encoded in this model use the steady state values from the full rate equations. Consequently, the response of the system to noise will not mimic that of the G&R model - especially when dramatic changes are introduced. The linewidth is a dynamic parameter, hence the transient behaviour of the system is not completely captured. However, the amplitude of transients are much smaller than those of, for example a chirped laser. The agreement with experimental values suggest that these effects do not render the model predictions invalid. From the previous section, the linewidths are also in line with experimental values. This suggests that the models approximate weightings of spontaneous emission and LEF are accurate when

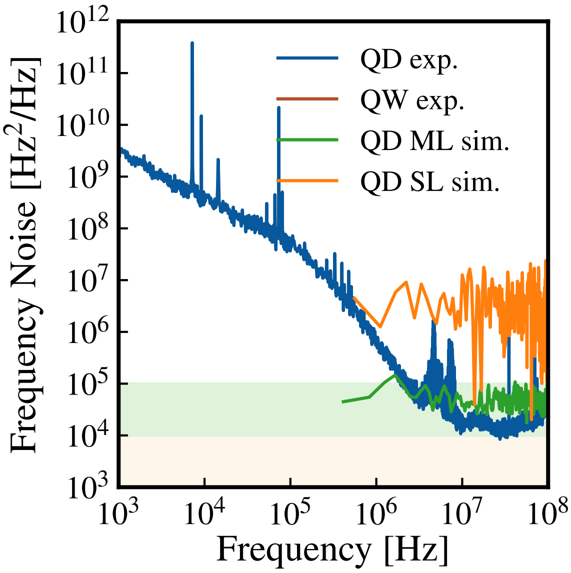

Figure 5.9: Frequency noise spectrum for free-running quantum dot lasers at 0dBm output power. Experimental data is from [9].

calculating the linewidth. Importantly, the free running linewidth calculated in the multi-level model is below the 200 kHz threshold. The next section will evaluate how the linewidth could be further reduced, which could increase the SKR.

5.3 Dynamics under EOF

The final section investigates the effect of locking on linewidth. A key parameter in the fibre/DBR setup used is the grating strength of the DBR, which determines the reflection strength and the linewidth. The linewidth was plotted against the feedback strength, which was controlled by changing the grating strength (κL) over a range from 0 to 2. This is shown in Figure 5.10 (a). The zones of locking can be observed here. In Zone II the linewidth steadily decreases with increasing feedback strength. The linewidth reached a minimum of 39 Hz at a feedback strength of -11.6 dB.

Compared to our linewidth value, the minimum linewidth found by Dong et al. under EOF is much reduced, reaching 0.6 Hz. A key consideration is the length of optical fibre cavity. In the PICWave simulations, the waveguide length was capped at 10 cm to reduce simulation time. Experimentally, a value of 4 m was used. The NRF scales with the length of the grating, according to Equation 3.1. Calculating the ratio of NRF using the paper parameters suggests

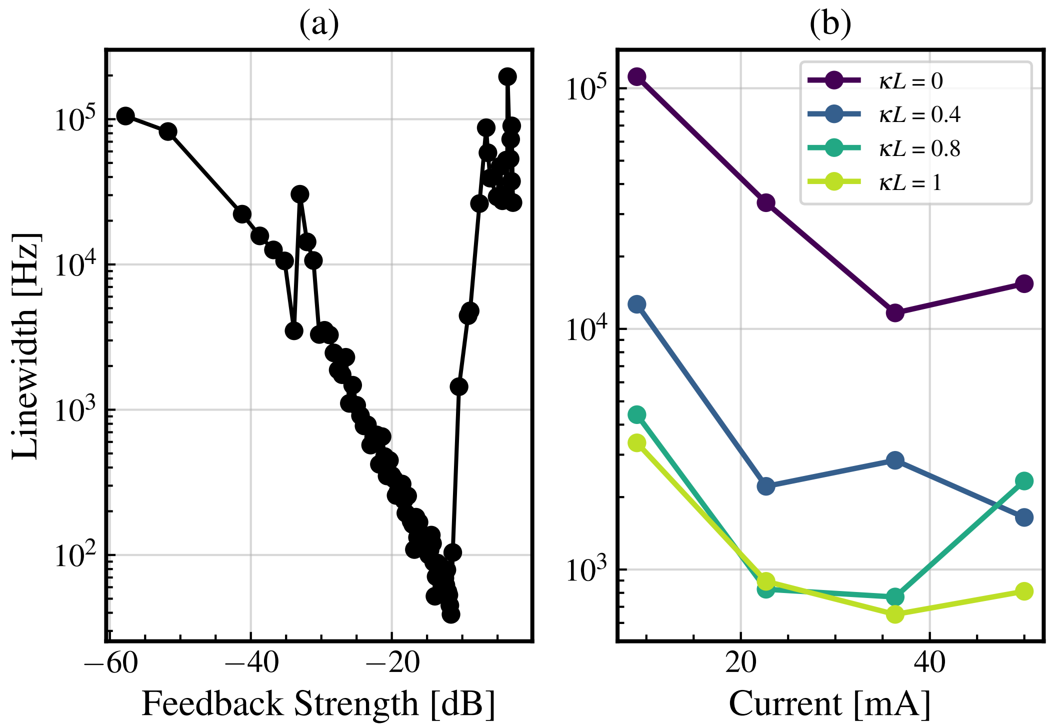

Figure 5.10: (a) Linewidth plotted against feedback strength, which is controlled by the grating strength. (b) Linewidth against current current as a function of grating strength.

that this accounts for a factor of ≈ 12 difference.

The coupling factor, which scales with the cavity refractive index, was also reduced by a factor of 1.5 by the erroneously specified waveguide index, as discussed in the methodology. This would have the effect of decreasing the intensity of the feedback at a given grating strength. Therefore a larger grating strength is required to provide the same level of feedback. The maximum linewidth reduction found in this work was at $\kappa L = 1.5$, compared to $\kappa L = 1$ for Dong et al. At a higher grating strength the Q factor is lower. At the same feedback strength, our Q factor will therefore be lower. Therefore coherence collapse will occur at a lower feedback power. As expected, the experimental maximum feedback tolerance is higher: -8.1 dB, compared to the -11 dB predicted by simulation.

Figure 5.10 (b) shows the behaviour of linewidth with current. As expected the linewidth decreases as current increases - power increases, which is inversely proportional to linewidth by Schawlow-Townes. The lowest linewidth is exhibited by the highest grating strength tested. The relation is not monotonic, which draws to attention another consideration - linewidth accuracy.

The measurement at $\kappa L = 0.8$ was repeated 5 times. The average linewidth was 1270 Hz with a standard deviation of 300 Hz. These results therefore indicate that the qualitative behaviour of the linewidth under EOF is effectively captured by the model, but that a more thorough interpretation of linewidth is necessary to make device predictions. For example, the frequency noise spectra for the different grating strengths could be compared for a longer duration, and the Lorentzian linewidth extracted from the white noise region.

However, we have demonstrated the flexibility of building a QD model in PICWave. An external locking circuit has been constructed, and linewidth narrowing has been observed. Although the linewidth values do not reach the same ultra-low levels, the refractive index and length of the external waveguide have been identified as possible reasons for this.

Chapter 6

Conclusion

CV-QKD relies on ultra-stable, narrow-linewidth lasers to prevent phase-drift. To satisfy CV-QKD security proofs, a maximum laser linewidth of 200 kHz is required, although further linewidth reduction can increase the secret key rate. In this work we have demonstrated a multi-level QD DFB laser model, whose qualitative behaviour closely matches literature findings. The key parameter, the (free-running) linewidth was calculated as 160 kHz, which falls within the experimental linewidths of 41 kHz - 243 kHz.

Three key characteristics of the model were defined in the introduction. The first was that the model should be accurate. The multi-level PICWave model showed the same qualitative trend population dynamics, but the optimisation step over parameters did dilute the significance of the physical parameters. For example, it erased one of the escape transitions in the cascade model. Despite this, the L-I characteristics agreed well with the G&R values, although gain compression was observed at high currents. Additionally, high damping was found from the RIN spectrum, which meant that the model could handle high-levels of optical feedback (-11 dB), with significant linewidth reduction to 39 Hz. However, the resonance frequency was overestimated, which appears to be a result of increased differential gain.

Encouragingly, different model parameters suggest that these flaws are not inherent to the model structure. This leads on to the second goal defined in the introduction: the model should be interpretable. A SHAP analysis of a GP surrogate model shows the influence of the fitted parameters on the laser characteristics. For example the carrier capture times strongly dictate the L-I traits, whilst spontaneous emission rates were less impactful. This means that the model parameters could be more deliberately chosen to encourage physical behaviour. Using the GPs predictions to choose model parameters could form the basis of future work, as it appears to accurately model the LI score, and to a lesser extent the population score. A composite cost function should be used to optimise for realistic occupation probabilities and L-I characteristics at the same time. Further improvements could add additional factors to the cost function - for example to reward smoothness of the population dynamics. The report also demonstrates interpretability in a different manner. A framework has been established to understand the interplay of factors contributing to linewidth, and behaviour under EOF. For example, a method to simulate the alpha parameter has been used, which calculates α_H as 0.34 - within the range of literature values. This validates that the model calculates the spontaneous emission rates to a reasonable value, since α_H depends on linewidth and spontaneous emission rate.

Although the trends shown by these time constants are insightful, the values themselves compensate for secondary effects caused by the steady state nature of the Pauli blocking factors. Therefore these values do not map one-to-one onto physical values. However, these fitting parameters do afford the model more flexibility - the third target for the model. Different systems could theoretically be implemented, and future work could investigate the gain spectra of different distributions of quantum dots, and the resulting performance on the laser. The flexibility of the model was also demonstrated with the EOF experiments, whereby the laser diode was incorporated in a larger photonic circuit. Ultimately, this model could be built upon to introduce different sources of noise. For example the current into the laser could be modulated to investigate flicker noise. Extensions could simulate electronic locking with the Pound-Drever-Hall method. Another important factor is temperature fluctuations. The Python model could be adapted to produce gain spectra as a function of carrier density and temperature.

Overall, the multi-level model captures the behaviour of QD laser dynamics in a way that cannot be modelled from a single-level model. The linewidth has demonstrated accuracy compared to experimental values. The implementation, which uses Python and PICWave, means that the model could be developed to investigate non-idealities of QD distribution, and different operating conditions, towards photonic integrated circuits for CV-QKD applications.

Chapter 7

Appendix

7.1 Risk assessment retrospective

A risk assessment was not relevant to this project as there was no practical component.

7.2 Correction to G&R rate equations

This note refers to the rate equations from [14].

Equation (5) should read:

$$\frac{dN^h_{\mathrm{SCH}}}{dt} = m_h \frac{J}{e} \Delta z W - \frac{N^h_{\mathrm{SCH}}}{\tau_c^{h,W}} + \frac{N^h_{\mathrm{QD}}}{\tau_e^h} - \frac{N_T^e}{\tau_r^{e,W}}. \tag{7.1}$$

The red term is omitted from the reference paper, but must be included such that the number of electrons and holes are equal.

Equation (7) contains a sign error. The below equation highlights the correction:

$$N^h_{\mathrm{QD}} = \sum_{m=\mathrm{GS,ES1,\ldots,ES4}} \frac{D_m N_D W \Delta z N_l}{1 + e^{(E_m^h - E_F^h)/(k_B T)}} + \frac{k_B T m_h^*}{\pi \hbar^2} \ln(1 + e^{(E_F^h - E^h_{\mathrm{WL}})/(k_B T)}) N_l W \Delta z \tag{7.2}$$

.

7.3 Parameters to calculate optical properties

The following parameters are not included in G&R's [14] simulation framework.

Symbol	Description	Values
	MATERIAL PARAMETERS	
$\tau_e^{e,w}$	electron diffusion time to WL	2.86 ps
τ_e^{e,ES_2}	electron relaxation time ES_2 to ES_1	59.6 ps
τ_e^{e,ES_1}	electron relaxation time ES_1 to GS	7.29 ps
$\tau_e^{e,\mathrm{GS}}$	electron relaxation time to GS	15.5 ps
m_h^{SCH}	hole effective mass in SCH	$0.51m_e$
m_e^{SCH}	electron effective mass in SCH	$0.063m_e$
m_h^{QD}	hole effective mass in QD	$0.40m_e$
m_e^{QD}	electron effective mass in QD	$0.03m_e$

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